



# UNIVERSITI TEKNIKAL MALAYSIA MELAKA

## FAKULTI TEKNOLOGI MAKLUMAT DAN KOMUNIKASI

### WORKSHOP 1

### REPORT

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**Program** : BITE S1G1

**Project Title** : Library Management System

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**Supervisor Signature** : \_\_\_\_\_

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BIN CHE PEE @ CHE HANAPI

## EXECUTIVE SUMMARY

The **Library Management System** is designed to help librarians manage their libraries more efficiently. This system simplifies tasks such as tracking book inventory, managing member records, and processing borrowing transactions. When a book title or ID is entered, the system provides details about the book, including its availability, author, publication date, and location in the library.

The system offers an intuitive and user-friendly interface, making it easy for librarians, staff, and members to navigate and use the platform. In large libraries, manually handling books and member records can be challenging. The Library Management System automates these processes, ensuring accurate and up-to-date information. It is updated in real-time whenever new books are added or borrowed.

When a book is borrowed or returned, the system automatically updates the records and calculates overdue fines if necessary. To develop this system, **Microsoft Visual Studio** with **C++** will be used for programming, and **MySQL** will serve as the database for storing all library-related data.

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## **CHAPTER 1: INTRODUCTION**

### **1.1 Introduction**

The Library Management System falls under the domain of Business & Services, specifically focusing on improving the efficiency and effectiveness of library operations. Libraries play a vital role in providing access to educational and informational resources, but many still rely on manual processes for managing their collections, member records, and transactions. These outdated methods often lead to inefficiencies, errors, and delays, which can negatively impact the quality of service provided to library patrons.

In today's fast-paced digital age, the need for automation in library management has become increasingly important. The Library Management System aims to address these challenges by automating key processes such as book inventory management, member registration, borrowing and returning transactions, and overdue fine calculations. By leveraging modern technologies like C++ for programming and MySQL for database management, the system provides a robust and scalable solution for libraries of all sizes.

The primary goal of this project is to create a system that not only simplifies the management process for librarians but also enhances the overall experience for library members. By automating routine tasks, the system reduces the administrative burden on library staff, allowing them to focus on providing better services to their patrons. Additionally, the system ensures real-time updates and accurate record-keeping, minimizing errors and discrepancies in book availability and member records.

In summary, the Library Management System represents a significant step forward in modernizing library operations. By automating key processes and providing a user-friendly interface, the system aims to improve operational efficiency, enhance data accuracy, and provide a better experience for both library staff and members.

## 1.2 Problem Statement

1. **Crowded Library Spaces:** During peak periods, such as exam seasons or special events, libraries often experience overcrowding. This makes it difficult for staff to manage book borrowing and returning processes efficiently, leading to long queues and delays in service.
2. **Inaccurate Book Tracking:** Manual tracking of book availability often results in errors, such as double bookings or lost materials. Without real-time updates, librarians cannot provide accurate information to members about the availability of books, leading to frustration and dissatisfaction.
3. **Delayed Updates on Borrowing Records:** Manually updating borrowing histories is time-consuming and prone to errors. Delays in updating member records can cause discrepancies, such as books being marked as available when they are still in possession of a member, or confusion over due dates.
4. **Inaccurate Overdue Fine Calculations:** Overdue fines are often calculated manually, which increases the risk of human error. Librarians may either overcharge or undercharge members, leading to disputes and dissatisfaction. Additionally, manual calculations are time-consuming, especially when managing a large volume of borrowed books.
5. **Time-Consuming Report Generation:** Generating reports manually, such as those for overdue books, borrowing trends, or popular titles, takes significant time and effort. This limits the ability of librarians to make informed decisions about inventory management or provide insights into the usage of the library's resources.

### **1.3 Objective (s) of the project**

This project embarks on the following objectives:

1. To investigate the current challenges faced by libraries in managing books, members, and transactions manually.
2. To design a Library Management System that automates the management of book inventory, member records, and borrowing transactions.
3. To conclude the development of the Library Management System by implementing features such as realtime data tracking, overdue fine calculations, and report generation.

### **1.4 Scope**

#### **1.4.1 Module to be developed**

##### **1. Authentication Module**

- Register
- Login
- Authentication
- Forgot Password

##### **2. User Management Module**

- Add / Edit / Delete User
- Search User

##### **3. Book Management Module**

- Add New Book
- Edit / Delete Book
- Search Book

##### **4. Borrowing and Returning Module**

- Borrow Book
- Return Book
- View Borrowing History

##### **5. Overdue and Fine Management Module**

- Track Overdue Books
- Calculate Fines
- Pay Fines

## **6. Reports and Analytics Module**

- Generate Borrowing Reports
- Generate Overdue Reports
- Generate Inventory Reports

## **7. Category Management Module**

- Add / Edit / Delete Category
- Search Category

## **8. Notification Module**

- Due Date Reminders
- Overdue Notifications
- New Book Alerts

## **9. Admin Dashboard Module**

- System Overview
- User Activity Tracking
- System Settings

## 1.4.2 Target User

### 1. Administrator

- Responsible for managing the system, staff, and library inventory.
  - Login
  - Manage Staff (CRUD)
  - Manage Book (CRUD)
  - Generate Reports (Borrowing, Overdue, Top Borrowers)
  - Logout

### 2. Staff / Librarian

- Responsible for day-to-day operations, including managing book inventory and assisting students.
  - Login
  - Assisting the Borrow and Return book process.
  - Manage Books (CRUD)
  - Manage Students (CRUD)
  - Logout

### 3. Student

- Students need to register to log in to the system to borrow books. They can view their borrowing history and pay fines if applicable.
  - Login
  - View Available Books
  - View Borrowed Books
  - Check Borrowing Limits
  - View Borrowing History
  - Check Fines
  - Logout

## **1.5 Project Significance**

### **1) Enhancing library management efficiency**

The library management system streamlines daily operations, reducing manual workload for librarians and staff. By automating tasks such as cataloging, circulation, and patron management, the system enables library personnel to focus on providing better services to users and optimizing resource allocation.

### **2) Improving user experience and accessibility**

The system offers a user-friendly interface that allows patrons to easily search for materials, check their borrowing status, and place reservations online. This improves accessibility by enabling users to interact with the library's resources remotely, saving time and effort. The system also facilitates personalized recommendations based on user preferences, enhancing their overall library experience.

### **3) Facilitating data-driven decision making**

The library management system generates comprehensive reports and analytics on circulation patterns, popular titles, and more. These insights enable library administrators to make data-driven decisions regarding collection development, resource allocation, and service improvements. By understanding user needs and preferences, libraries can optimize their offerings and better serve their communities.

### **4) Contributing to the digital transformation of libraries**

Implementing a library management system is a significant step towards the digital transformation of libraries. It prepares libraries to adapt to the changing needs of users in an increasingly digital world. By embracing technology and automating processes, libraries can enhance their relevance, attract new users, and position themselves as essential hubs for knowledge access and community engagement in the digital age.



## **CHAPTER 2: ANALYSIS OF PROBLEM**

### **2.1 Problem Decomposition Description**

#### **1) Manual Book Tracking**

In many libraries, book inventory is manually managed, often leading to significant issues such as errors in availability tracking, discrepancies in book status, and inefficient retrieval times. Libraries using manual systems can face difficulties ensuring real-time updates, resulting in double bookings, misplaced books, or slow responses to patron requests. The lack of automation significantly impacts both the user experience and library staff productivity.

#### **2) Outdated Member Records**

Manual management of library member information, including their borrowing history and overdue fines, often results in data inaccuracies. Without an automated system, member records can easily become outdated, leading to confusion over borrowing rights, overdue penalties, and lost or incorrect member contact details. This system not only makes managing large user bases difficult but also diminishes the overall reliability of library services.

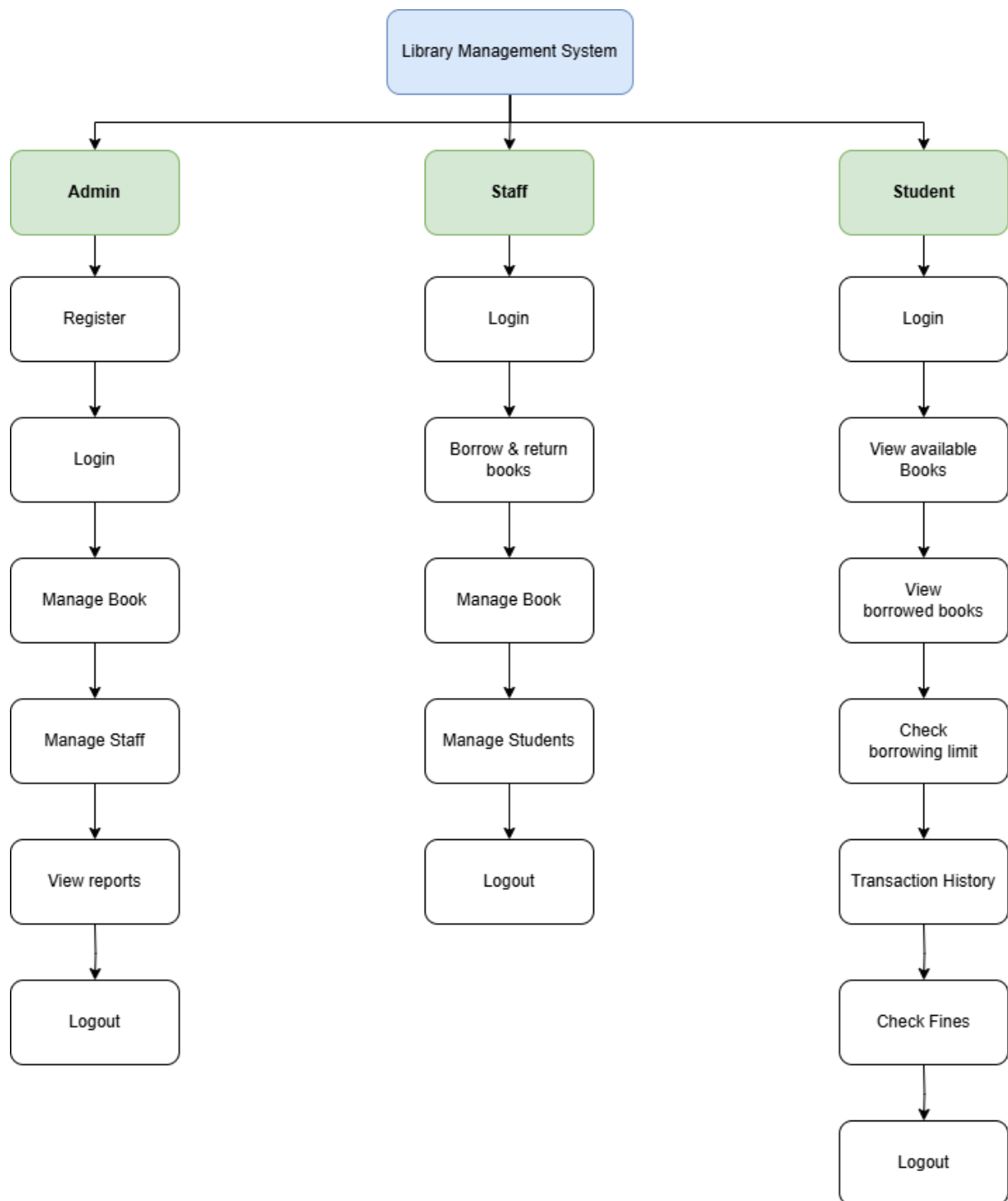
#### **3) Inefficient Fine Calculations**

Currently, libraries often struggle with calculating overdue fines manually, which can be time-consuming and prone to error. Without an automated method for calculating overdue fines, discrepancies can occur, leading to unfair penalties or lost revenue for the library. The manual tracking of fine statuses, combined with the irregular updating of records, can negatively affect both members and staff, causing dissatisfaction and operational inefficiencies.

#### **4) Report Generation Challenges**

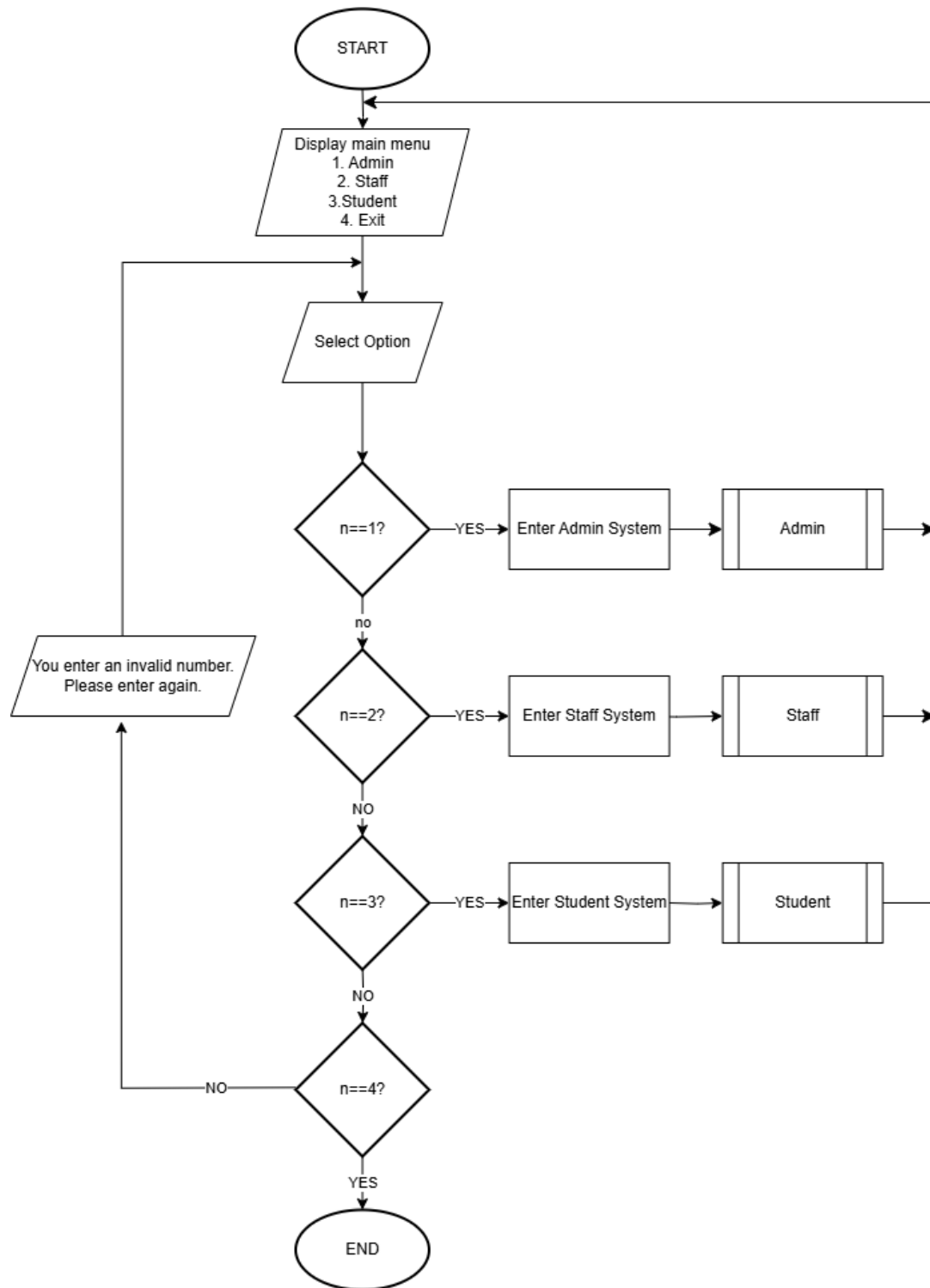
Generating accurate, up-to-date reports manually is both time-intensive and prone to error. Library staff often face delays in producing reports related to borrowing trends, overdue books, or inventory status. Without an automated system for generating these reports, libraries miss opportunities to use data effectively for decision-making. The lack of detailed and timely reports hampers strategic planning, resource allocation, and overall library management.

## 2.2 Structured Chart

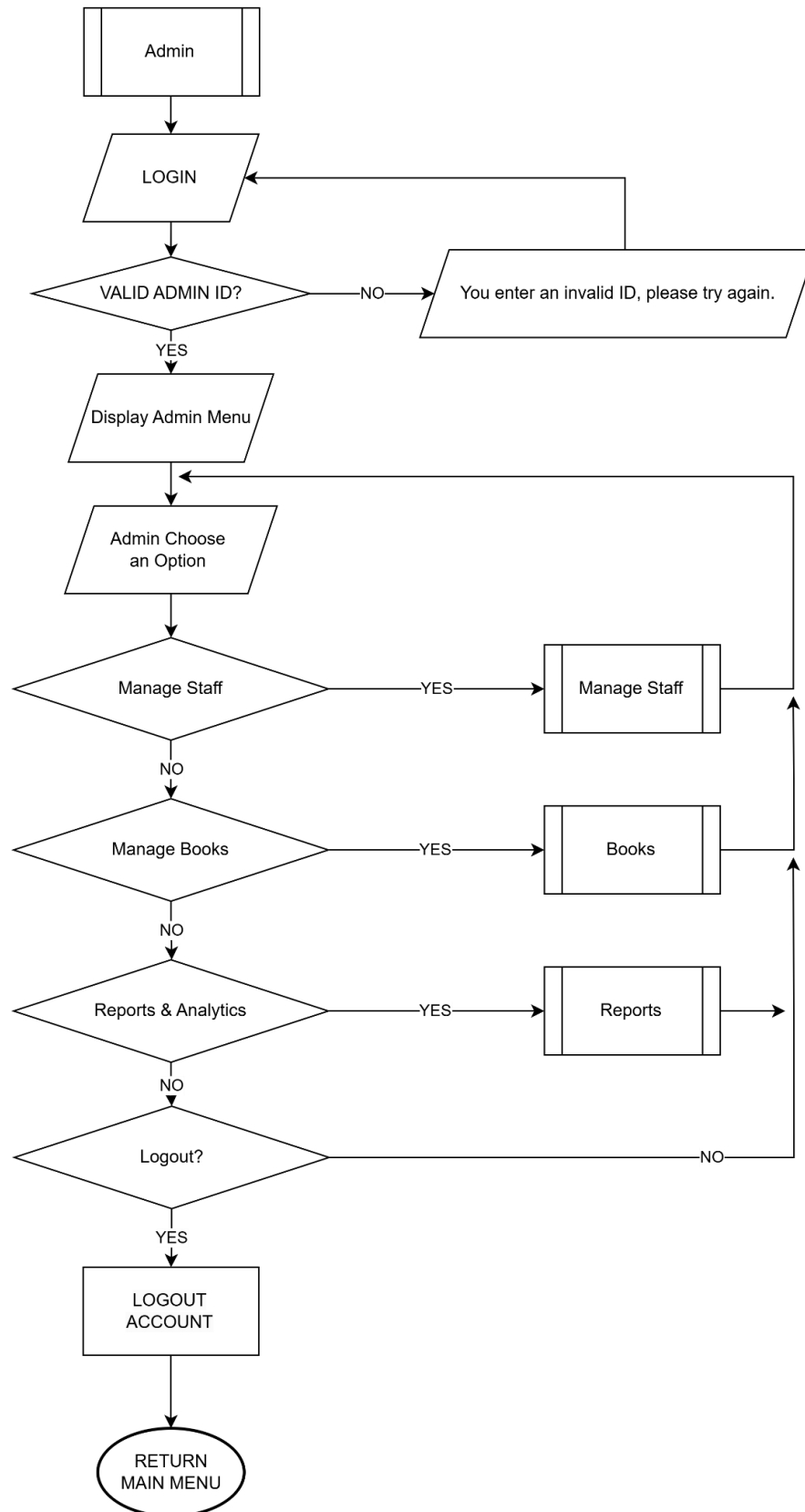


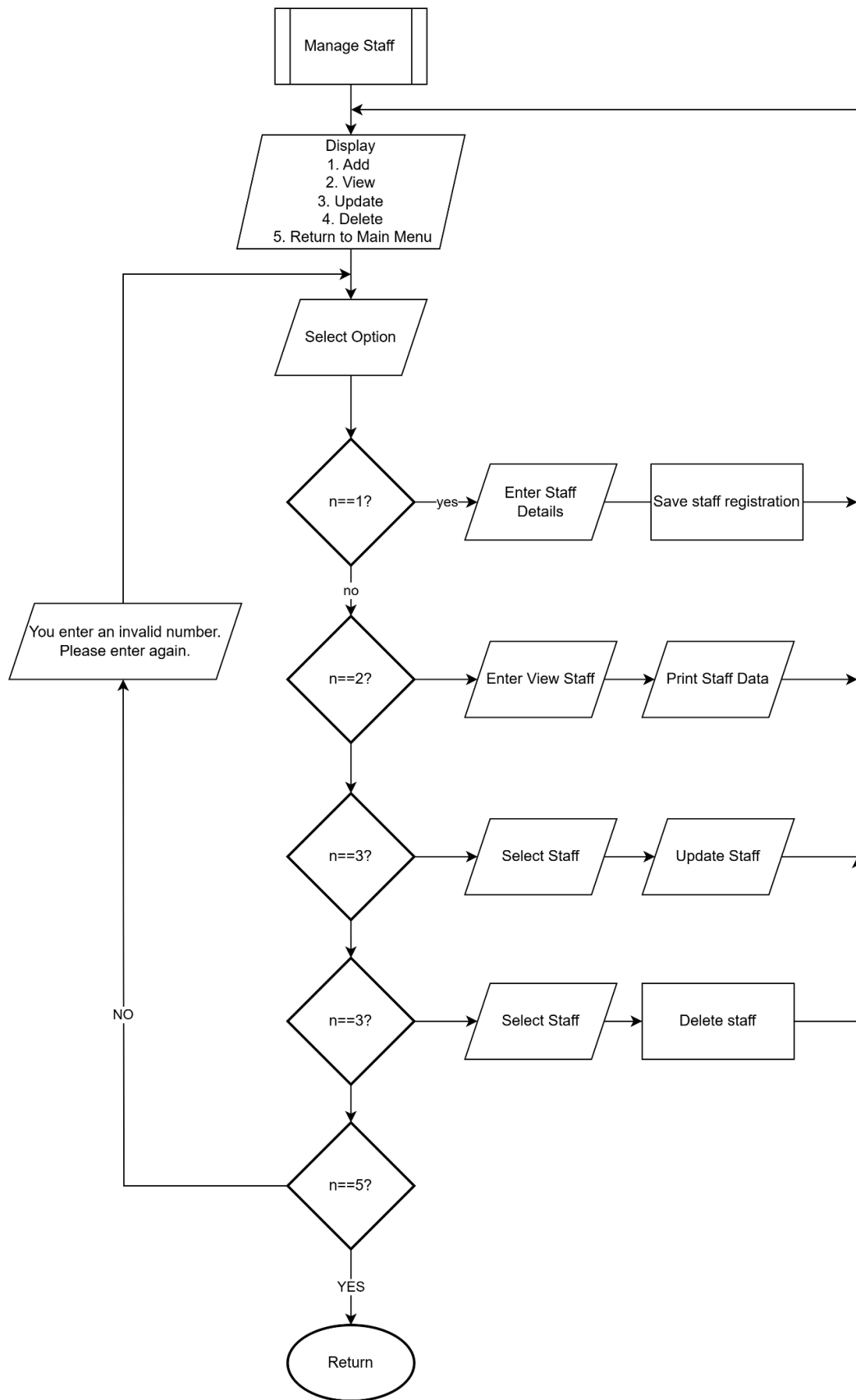
## CHAPTER 3: DESIGN

### 3.1 Flowchart

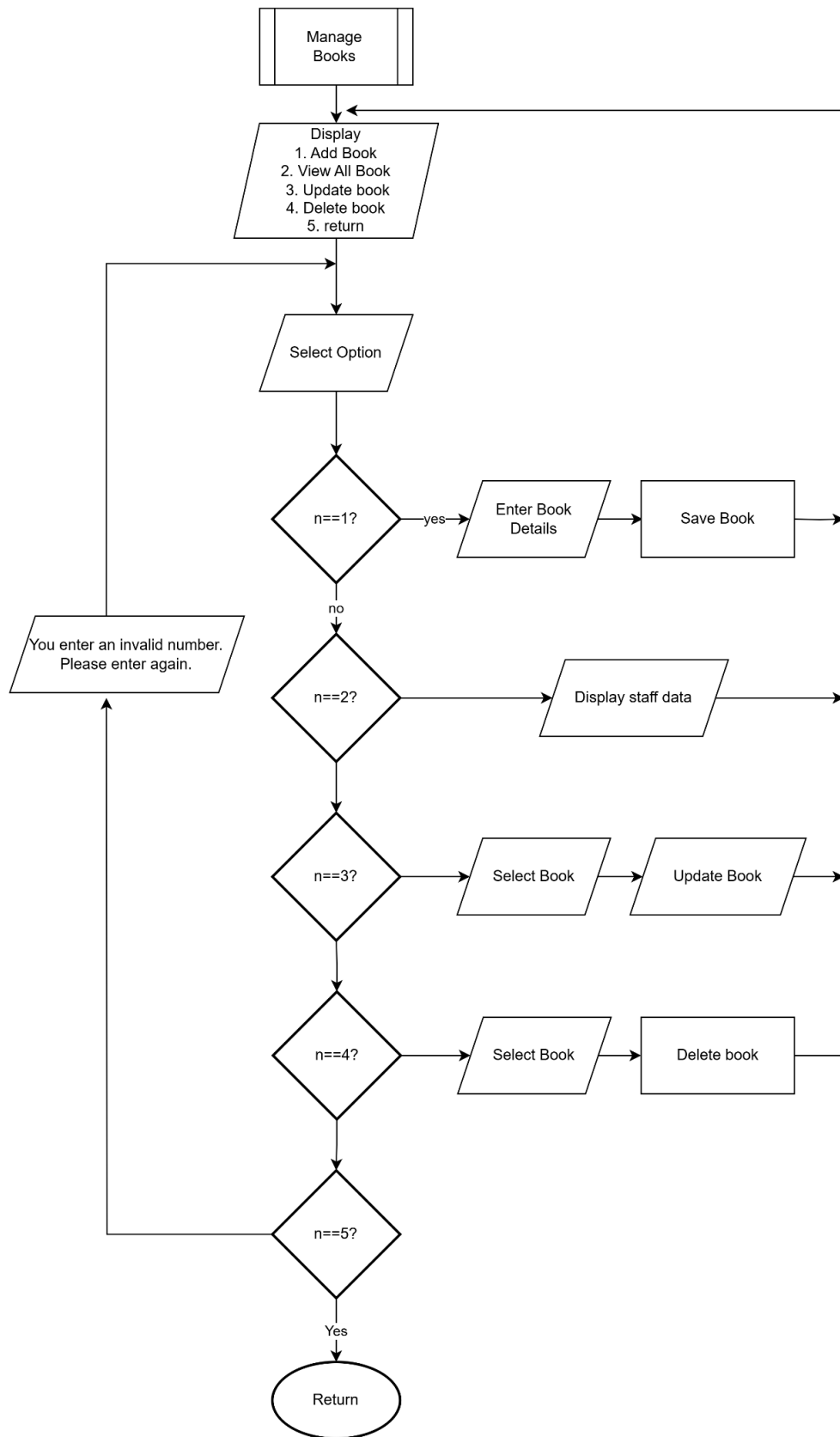


MAIN MENU

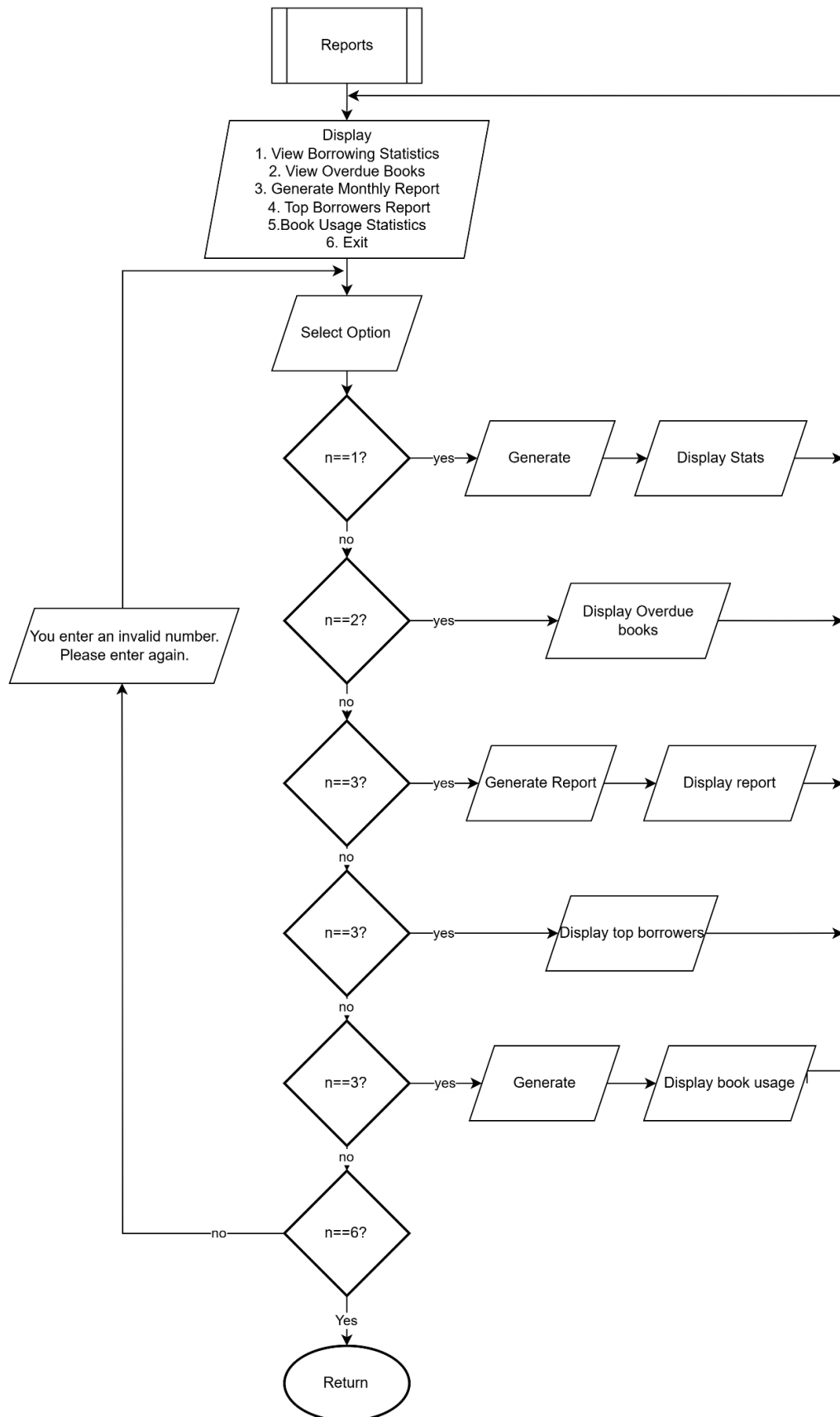
**ADMIN FLOWCHART**



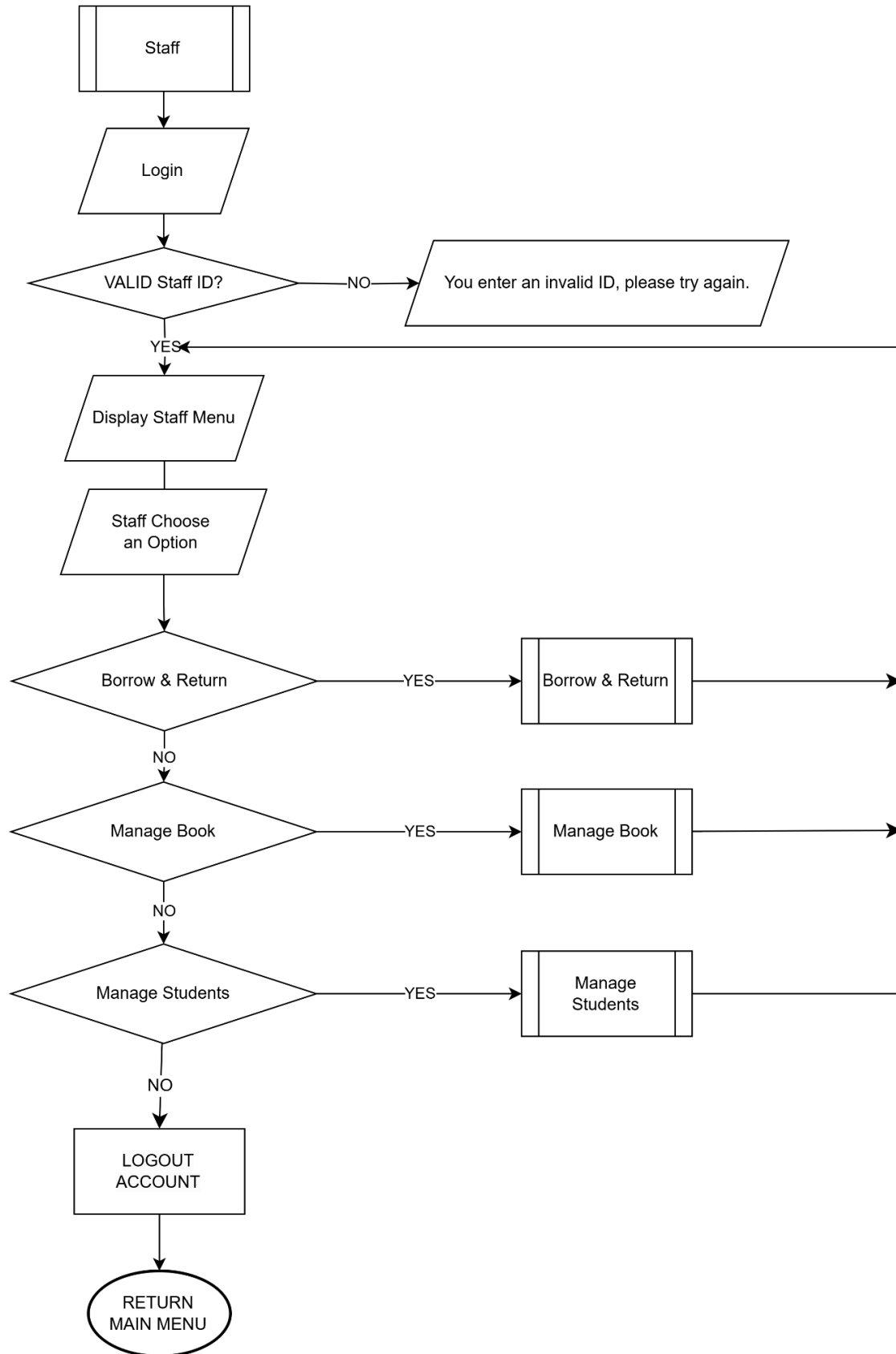
**MANAGE STAFF FLOWCHART**

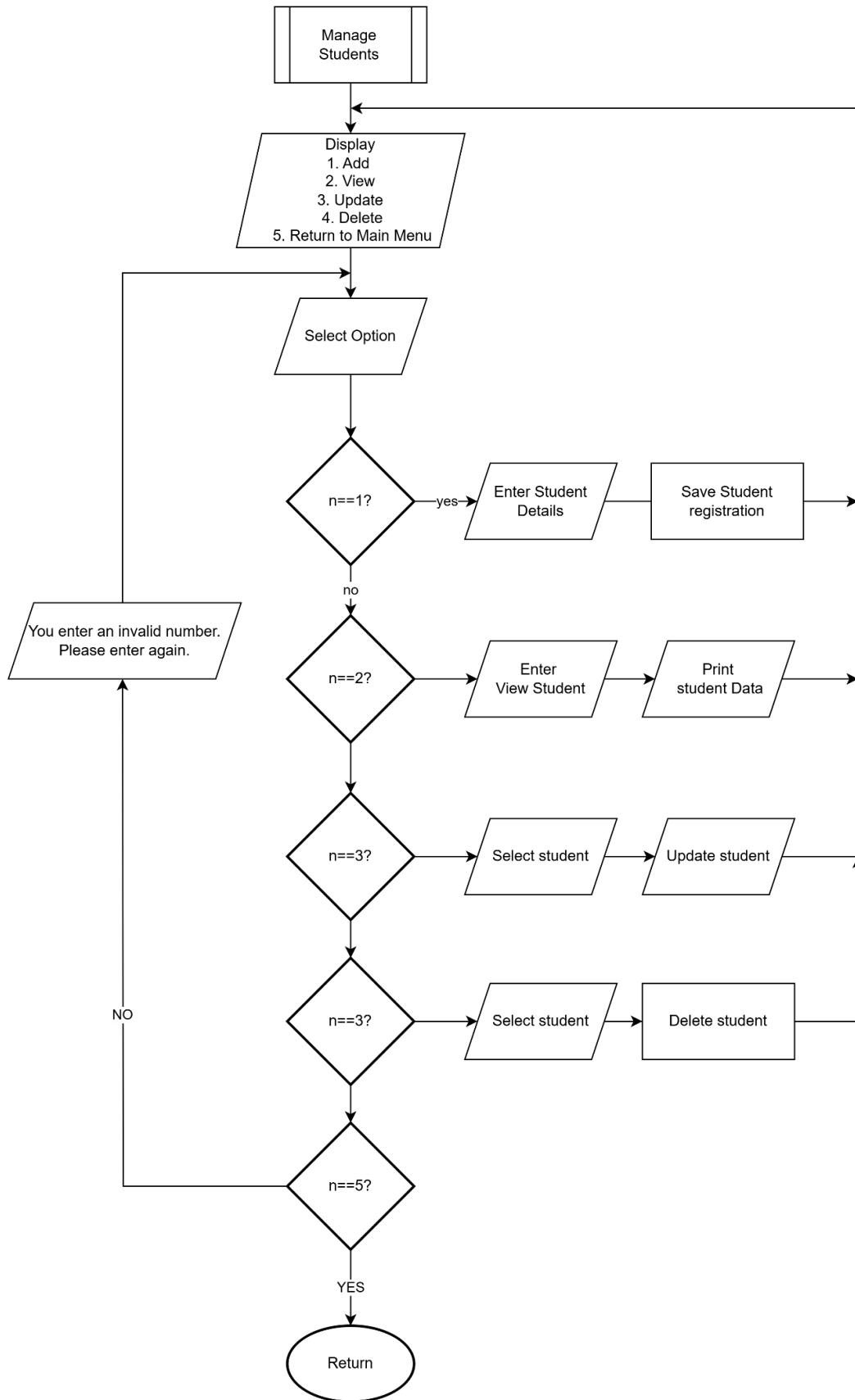


**MANAGE BOOKS FLOWCHART**

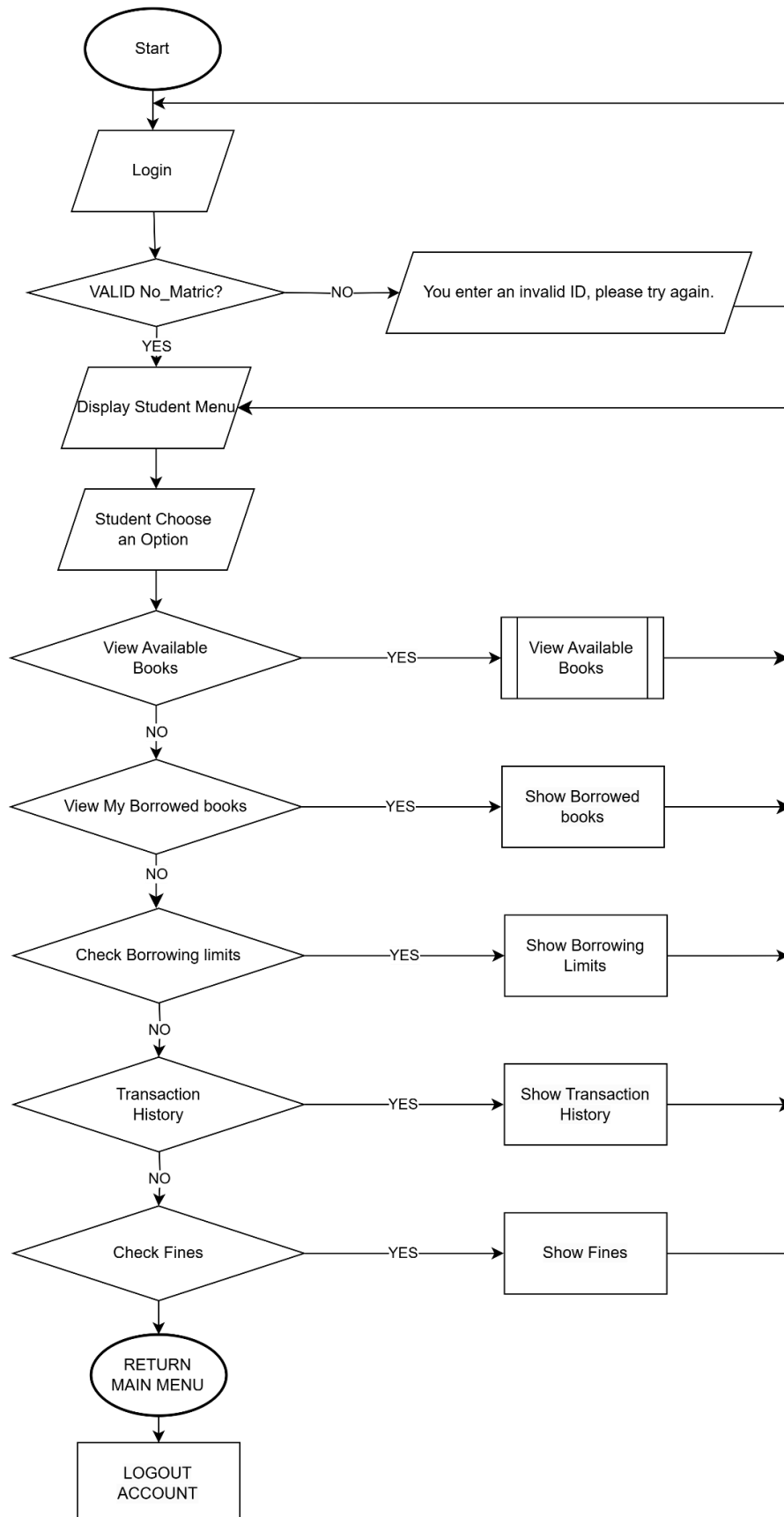


**REPORTS FLOWCHART**

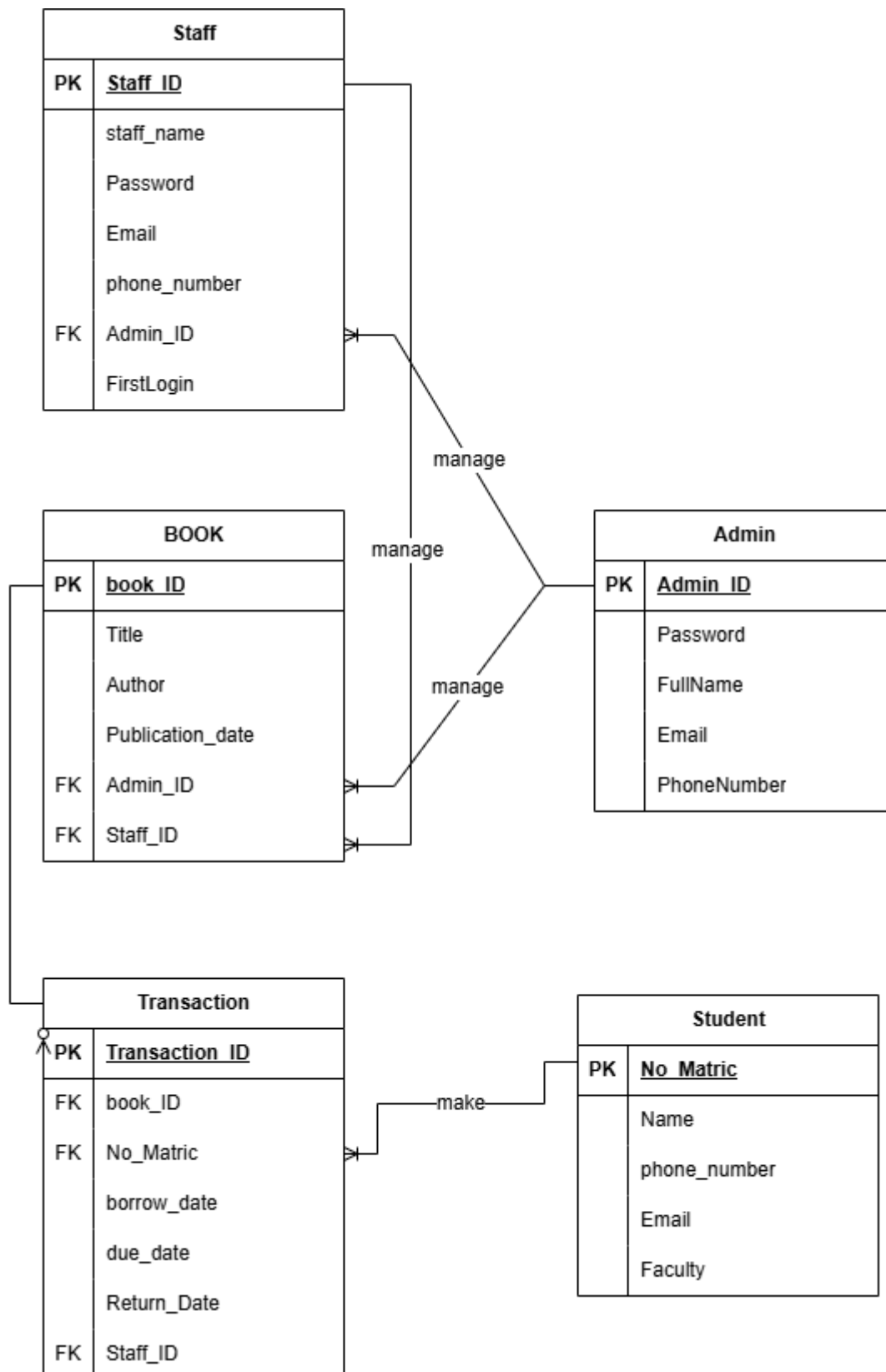
**STAFF FLOWCHART**



**MANAGE STUDENTS FLOWCHART**

**STUDENTS FLOWCHART**

### 3.2 ERD



### 3.3 Data Dictionary

#### ADMIN TABLE

| ATTRIBUTE NAME     | CONTENTS                | DATA TYPE AND SIZE | FORMAT       | RANGE            | REQUIRED? | DEFAULT VALUE | UNIQUE? | PK OR FK | FK REFERENCED TABLE |
|--------------------|-------------------------|--------------------|--------------|------------------|-----------|---------------|---------|----------|---------------------|
| <i>Admin_ID</i>    | Unique admin identifier | varchar(10)        | Alphanumeric | 1-10 characters  | Yes       | None          | Yes     | PK       | None                |
| <i>Password</i>    | Admin's password        | varchar(255)       | Alphanumeric | 1-255 characters | Yes       | None          | No      | None     | None                |
| <i>FullName</i>    | Admin's full name       | varchar(100)       | Text         | 1-100 characters | Yes       | None          | No      | None     | None                |
| <i>Email</i>       | Admin's email address   | varchar(100)       | Email        | 1-100 characters | Yes       | None          | Yes     | None     | None                |
| <i>PhoneNumber</i> | Admin's phone number    | varchar(15)        | Numeric      | 1-15 digits      | No        | None          | No      | None     | None                |

#### STAFF TABLE

| ATTRIBUTE NAME     | CONTENTS                   | DATA TYPE AND SIZE | FORMAT       | RANGE            | REQUIRED? | DEFAULT VALUE | UNIQUE? | PK OR FK | FK REFERENCED TABLE |
|--------------------|----------------------------|--------------------|--------------|------------------|-----------|---------------|---------|----------|---------------------|
| <i>Staff_ID</i>    | Unique staff identifier    | varchar(10)        | Alphanumeric | 1-10 characters  | Yes       | None          | Yes     | PK       | None                |
| <i>FullName</i>    | Staff's full name          | varchar(100)       | Text         | 1-100 characters | Yes       | None          | No      | None     | None                |
| <i>Email</i>       | Staff's email address      | varchar(100)       | Email        | 1-100 characters | Yes       | None          | Yes     | None     | None                |
| <i>PhoneNumber</i> | Staff's phone number       | varchar(15)        | Numeric      | 1-15 digits      | No        | None          | No      | None     | None                |
| <i>Role</i>        | Staff's role in the system | varchar(50)        | Text         | 1-50 characters  | Yes       | None          | No      | None     | None                |

#### STUDENT TABLE

| ATTRIBUTE NAME     | CONTENTS                  | DATA TYPE AND SIZE | FORMAT       | RANGE            | REQUIRED? | DEFAULT VALUE | UNIQUE? | PK OR FK | FK REFERENCED TABLE |
|--------------------|---------------------------|--------------------|--------------|------------------|-----------|---------------|---------|----------|---------------------|
| <i>Student_ID</i>  | Unique student identifier | varchar(10)        | Alphanumeric | 1-10 characters  | Yes       | None          | Yes     | PK       | None                |
| <i>FullName</i>    | Student's full name       | varchar(100)       | Text         | 1-100 characters | Yes       | None          | No      | None     | None                |
| <i>Email</i>       | Student's email address   | varchar(100)       | Email        | 1-100 characters | Yes       | None          | Yes     | None     | None                |
| <i>PhoneNumber</i> | Student's phone number    | varchar(15)        | Numeric      | 1-15 digits      | No        | None          | No      | None     | None                |

|                       |                    |      |      |            |     |              |    |      |      |
|-----------------------|--------------------|------|------|------------|-----|--------------|----|------|------|
| <i>EnrollmentDate</i> | Date of enrollment | DATE | Date | YYYY-MM-DD | Yes | CURRENT_DATE | No | None | None |
|-----------------------|--------------------|------|------|------------|-----|--------------|----|------|------|

## BOOK TABLE

| ATTRIBUTE NAME  | CONTENTS                   | DATA TYPE AND SIZE | FORMAT       | RANGE            | REQUIRED? | DEFAULT VALUE | UNIQUE? | PK OR FK | FK REFERENCED TABLE |
|-----------------|----------------------------|--------------------|--------------|------------------|-----------|---------------|---------|----------|---------------------|
| <i>Book_ID</i>  | Unique book identifier     | varchar(13)        | Alphanumeric | 1-13 characters  | Yes       | None          | Yes     | PK       | None                |
| <i>Title</i>    | Book's title               | varchar(255)       | Text         | 1-255 characters | Yes       | None          | No      | None     | None                |
| <i>Author</i>   | Author(s) of the book      | varchar(255)       | Text         | 1-255 characters | Yes       | None          | No      | None     | None                |
| <i>ISBN</i>     | Book ISBN                  | varchar(20)        | Alphanumeric | 1-20 characters  | Yes       | None          | Yes     | None     | None                |
| <i>Category</i> | Book's category            | varchar(50)        | Text         | 1-50 characters  | Yes       | None          | No      | None     | None                |
| <i>Quantity</i> | Number of available copies | INT                | Integer      | 0-9999           | Yes       | 0             | No      | None     | None                |

## TRANSACTION TABLE

| ATTRIBUTE NAME        | CONTENTS                  | DATA TYPE AND SIZE | FORMAT       | RANGE           | REQUIRED? | DEFAULT VALUE | UNIQUE? | PK OR FK | FK REFERENCED TABLE |
|-----------------------|---------------------------|--------------------|--------------|-----------------|-----------|---------------|---------|----------|---------------------|
| <i>Transaction_ID</i> | Unique transaction ID     | INT                | Integer      | 1-999999        | Yes       | None          | Yes     | PK       | None                |
| <i>Book_ID</i>        | Book ID borrowed          | varchar(13)        | Alphanumeric | 1-13 characters | Yes       | None          | No      | FK       | book                |
| <i>Student_ID</i>     | Student who borrowed book | varchar(10)        | Alphanumeric | 1-10 characters | Yes       | None          | No      | FK       | student             |
| <i>Borrow_Date</i>    | Date book was borrowed    | DATE               | Date         | YYYY-MM-DD      | Yes       | CURRENT_DATE  | No      | None     | None                |
| <i>Return_Date</i>    | Date book was returned    | DATE               | Date         | YYYY-MM-DD      | No        | NULL          | No      | None     | None                |
| <i>Fine</i>           | Fine for overdue books    | DECIMAL(10, 2)     | Numeric      | 0.00-9999.99    | No        | 0             | No      | None     | None                |

### 3.4 Interface Design

#### ADMIN

#### Register Admin

```
C:\Users\ACER\source\repos\ x + v

=====
Library Management System
=====
Welcome! The system requires an admin.
Please register your admin details to proceed.
=====

Step 1: Enter Admin ID (e.g., ADM001):> ADM001

Password Strength: ||||| 100%
(Press TAB to show/hide password)
Enter Password: > *****

Step 3: Enter Full Name: > Hamua bin Nurdin

Step 4: Enter Email: > hamua10@gmail.com

Step 5: Enter Phone Number: > 0102193549
```

#### MAIN MENU

```
C:\Users\ACER\source\repos\ x + v

=====
Library Management System
Main Menu
=====

1. Admin
2. Staff
3. Student
4. Exit

=====
Enter your choice: |
```

```
=====
                        Admin Login
=====

Enter Admin ID:          > ADM001

(Press TAB to show/hide password)
Enter Password:          > *****|
```

## ADMIN LOGIN

```
=====
                        ADMIN MAIN MENU
=====

Welcome, Hamua bin Nurdin!

1. Staff Management
2. Book Management
3. Reports & Analytics
4. Logout

=====
Enter your choice:
```

## ADMIN MENU

## ADMIN MENU - STAFF MANAGEMENT MENU

```
=====
                        STAFF MANAGEMENT MENU
=====

1. Add Staff
2. View All Staff
3. Update Staff
4. Delete Staff
5. Return to Main Menu

=====
Enter your choice: |
```

```
=====
                        ADD NEW STAFF
=====

Enter Staff ID (e.g., STF001) or 'B' to go back: > STF001

Enter Staff Name:           > Nur Aina Haura

Enter Email:                > nurainahaura27@gmail.com

Enter Phone Number:        > 0106007834

Password Strength: ||||| 100%
(Press TAB to show/hide password)
Enter Password:            > Stff001!
```

## ADMIN MENU – STAFF MANAGEMENT(ADD)

```
=====
                        CONFIRM STAFF DETAILS
=====

Staff ID: STF002
Name: Meysya El-zahra
Email: meysya10@yahoo.com
Phone: 0156234982

=====
Confirm adding this staff? (Y/N): > y

Staff added successfully!

Press any key to continue...
```

## ADMIN MENU – STAFF MANAGEMENT(ADD CONFIRMATION)

| STAFF RECORDS                                          |                                   |                          |            |             |
|--------------------------------------------------------|-----------------------------------|--------------------------|------------|-------------|
| Total Records: 5   Sort by: Staff_ID ASC   Page 1 of 1 |                                   |                          |            |             |
| Staff ID                                               | Staff Name                        | Email                    | Phone      | First Login |
| STF001                                                 | Nur Aina Haura                    | nurainahaura27@gmail.com | 0106007834 | Yes         |
| STF002                                                 | Meysya EL-zahra                   | meysya10@yahoo.com       | 0156234982 | Yes         |
| STF003                                                 | Mohd Salehudin bin Iwan           | saleh47@gmail.com        | 0146322965 | Yes         |
| STF004                                                 | Andi Mohd Nursyakirman            | andikiman29@gmail.com    | 0174562196 | Yes         |
| STF005                                                 | Nurul Khuzaimah Natasya bt Samade | nurulkhuza28@gmail.com   | 0149144297 | Yes         |

Options:  
 [</>] Previous/Next Page  
 [S] Search [O] Sort [R] Reset Filters  
 [B] Back to Menu  
 Enter choice: |

### ADMIN MENU – STAFF MANAGEMENT (VIEW)

| DELETE STAFF |                                   |                          |            |             |
|--------------|-----------------------------------|--------------------------|------------|-------------|
| Staff ID     | Staff Name                        | Email                    | Phone      | First Login |
| STF001       | Nur Aina Haura                    | nurainahaura27@gmail.com | 0106007834 | Yes         |
| STF002       | Meysya EL-zahra                   | meysya10@yahoo.com       | 0156234982 | Yes         |
| STF003       | Mohd Salehudin bin Iwan           | saleh47@gmail.com        | 0146322965 | Yes         |
| STF004       | Andi Mohd Nursyakirman            | andikiman29@gmail.com    | 0174562196 | Yes         |
| STF005       | Nurul Khuzaimah Natasya bt Samade | nurulkhuza28@gmail.com   | 0149144297 | Yes         |

Enter Staff ID to delete (or 'B' to go back):> STF002

Staff Details  
 Staff ID: STF002  
 Name: Meysya EL-zahra

Are you sure you want to delete this staff? (Y/N):> n

Deletion cancelled.

Press any key to continue...

### ADMIN MENU – STAFF MANAGEMENT(DELETE)

## BOOK MANAGEMENT MENU

```
=====
                        BOOK MANAGEMENT MENU
=====

1. Add New Book
2. View All Books
3. Update Book
4. Delete Book
5. Return to Main Menu

=====
Enter your choice: |
```

```
=====
                        ADD NEW BOOK
=====

Enter Book Title (or 'B' to go back): > MATLAB
Enter Author Name: > Ozgun
Enter ISBN (13 digits): > 9781498784078
Enter Classification Code (e.g., QA, BP): > QA
Enter Subject Number (e.g., 76.3): > 297
Enter Cutter Number (e.g., .T39): > .093
Enter Publication Year (or 0 if unknown): > 2019
Enter Collection Symbol (e.g., os, ref): > os
Enter Floor Number (1-3): > 1
Enter Quantity (1-10): > 1|
```

## ADMIN MENU – BOOK MANAGEMENT (ADD)

```
=====
                        CONFIRM BOOK DETAILS
=====

Title: MATLAB
Author: Ozgun
ISBN: 9781498784078
Call Number: QA 297 .093 2019 os
Floor: 1
Quantity: 1
=====
Confirm adding this book? (Y/N): > y

Book added successfully!

Press any key to continue...
```

## ADMIN MENU – BOOK MANAGEMENT (ADD CONFIRMATION)

| BOOK CATALOG                                        |                      |                          |                          |       |     |
|-----------------------------------------------------|----------------------|--------------------------|--------------------------|-------|-----|
| Total Records: 8   Sort by: Title ASC   Page 1 of 2 |                      |                          |                          |       |     |
| ISBN                                                | Title                | Author                   | Call Number              | Class | Qty |
| 9789674816513                                       | Berani Speaking      | Coach Ridz               | PE 297.571 .R539 2021 os | PE    | 1   |
| 9789672145899                                       | Game Development     | Ibrahim Ahmad            | QA 75.5 .I27 os          | QA    | 0   |
| 9781498784078                                       | MATLAB               | Ozgun                    | QA 297 .093 2019 os      | QA    | 1   |
| 9789674480646                                       | Merindu Baginda Nabi | Habiburrahman El Shirazy | BP 5055.9 .H32 2019 n    | BP    | 0   |
| 0764507370                                          | SQL for Dummies      | Allen G. Taylor          | QA 76.3 .T39 2001 os     | QA    | 3   |

Options:  
 [</>] Previous/Next Page  
 [S] Search [O] Sort [R] Reset Filters  
 [B] Back to Menu  
 Enter choice: |

## ADMIN MENU – BOOK MANAGEMENT (VIEW)

| UPDATE BOOK INFORMATION       |                          |               |                          |          |  |
|-------------------------------|--------------------------|---------------|--------------------------|----------|--|
| Title                         | Author                   | ISBN          | Call Number              | Quantity |  |
| SQL for Dummies               | Allen G. Taylor          | 0764507370    | QA 76.3 .T39 2001 os     | 3        |  |
| MATLAB                        | Ozgun                    | 9781498784078 | QA 297 .093 2019 os      | 1        |  |
| Surrounded by Idiots          | Thomas Erikson           | 9789670015873 | BF 789.4 .E73 2019 os    | 1        |  |
| Teruntuk Langit Yang Bercanda | jiwa                     | 9789671692011 | BP 630.2 .J59 2022 os    | 1        |  |
| Game Development              | Ibrahim Ahmad            | 9789672145899 | QA 75.5 .I27 os          | 0        |  |
| The Art of Letting God        | Mizi Wahid               | 9789672459231 | BP 630.1 .W12 2021 os    | 1        |  |
| Merindu Baginda Nabi          | Habiburrahman El Shirazy | 9789674480646 | BP 5055.9 .H32 2019 n    | 0        |  |
| Berani Speaking               | Coach Ridz               | 9789674816513 | PE 297.571 .R539 2021 os | 1        |  |

Enter ISBN to update (or 'B' to go back): > 9781498784078

Current Title: MATLAB

Enter New Title (press Enter to keep current): > MATLAB-electromagnetic modeling

Current Author: Ozgun

Enter New Author (press Enter to keep current): >

Current ISBN: 9781498784078

Enter New ISBN (13 digits, press Enter to keep current): >

Current Call Number: QA 297 .093 2019 os

Enter New Call Number (press Enter to keep current): >

Current Quantity: 1

Enter New Quantity (must be a positive integer, press Enter to keep current): > |

## ADMIN MENU – BOOK MANAGEMENT (UPDATE)

```

=====
CONFIRM UPDATE
=====
ISBN: 9781498784078
New Title: MATLAB-electromagnetic modeling
New Author: Ozgun
New ISBN: 9781498784078
New Call Number: QA 297 .093 2019 os
New Quantity: 1
=====
Confirm update? (Y/N):      > y
Book information updated successfully!
Press any key to continue...

```

### ADMIN MENU – BOOK MANAGEMENT (UPDATE CONFIRMATION)

```

=====
DELETE BOOK
=====
Enter ISBN to delete (13 digits) or 'B' to go back: > 9781498784078
Are you sure you want to delete this book? (Y/N): > y
Book deleted successfully!
Press any key to continue...

```

### ADMIN MENU – BOOK MANAGEMENT (DELETE)

## REPORTS AND ANALYTICS

```

=====
REPORTS & ANALYTICS
=====
1. View Borrowing Statistics
2. View Overdue Books
3. Generate Monthly Report
4. Top Borrowers Report
5. Book Usage Statistics
6. Return to Main Menu
=====
Enter your choice: |

```

## STAFF MENU

```
=====
                        Staff Login
=====

Enter Staff ID:          > STF001

(Press TAB to show/hide password)
Enter Password:          > *****|
```

## STAFF LOGIN

```
=====
                        Staff Login
=====

First login detected. Please reset your password.

(Min 8 chars, 1 uppercase, 1 lowercase, 1 digit, 1 special)

Password Strength: ||||| 100%
(Press TAB to show/hide password)
Enter Password:          > Nuraina2004!

(Press TAB to show/hide password)
Enter Password:          > Nuraina2004!
Password reset successfully! Please log in with your new password.
```

## RESET PASSWORD FOR FIRST TIME LOGIN

```
=====
                        STAFF MENU
=====

1. Borrow & Return Books
2. Manage Books
3. Manage Students
4. Logout

=====
Enter your choice: |
```

## STAFF MENU

```
=====
BORROW & RETURN BOOKS
=====
1. Borrow Book
2. Return Book
3. Back to Main Menu
=====
Enter your choice: |
```

### STAFF MENU – BORROW & RETURN BOOKS

```
=====
BORROW A BOOK
=====
Enter Student Matric Number (or 'B' to go back): > B032310741|
```

### STAFF MENU – BORROW & RETURN BOOKS (BORROW BOOKS), ENTER STUDENT MATRIC NUMBER

```
=====
STUDENT INFORMATION
=====
Name: Riswan bin Hamua
Faculty: FTMK
Current Borrowed Books: 0
=====
Enter Book ISBN (or 'B' to go back): > |
```

### STAFF MENU – BORROW & RETURN BOOKS(ENTER BOOK ISBN)

```
=====
CONFIRM BORROWING
=====
Book Title: MATLAB
Author: Ozgun
Available Copies: 1
=====
Confirm borrowing? (Y/N): > y

Book borrowed successfully!
Please return the book within 14 days to avoid fines.

Press any key to continue...
```

### STAFF MENU – BORROW & RETURN BOOKS (BORROW BOOK CINFIRMATION)

```
=====
                        RETURN A BOOK
=====

Enter Student Matric Number (or 'B' to go back): > B032310741|
```

### STAFF MENU – BORROW & RETURN BOOKS (RETURN BOOKS), ENTER STUDENT MATRIC NUMBER

```
=====
                        RETURN A BOOK
=====

Enter Student Matric Number (or 'B' to go back): > B032310741
Student Name: Riswan bin Hamua

=====
| No   | Title           | ISBN           | Borrow Date   | Due Date      | Status   | Fine (RM) |
=====
| 1    | MATLAB          | 9781498784078  | 2025-01-20    | 2025-02-03    | Borrowed | 0.00      |
=====

Enter book number to return (1-1): > |
```

### STAFF MENU – BORROW & RETURN BOOKS(ENTER BOOK NUMBER)

```
=====
                        CONFIRM BOOK RETURN
=====

Student: Riswan bin Hamua
Book Title: MATLAB
ISBN: 9781498784078
Due Date: 2025-02-03
Book returned on time - No fine

=====
Confirm return? (Y/N):      > y
=====

Book returned successfully!

Press any key to continue...
```

### STAFF MENU – BORROW & RETURN BOOKS (RETURN BOOK CINFIRMATION)

```

=====
MANAGE STUDENTS
=====

1. Add New Student
2. View All Students
3. Update Student
4. Delete Student
5. Return to Main Menu

=====
Enter your choice: |

```

## STAFF MENU – MANAGE STUDENT MENU

```

=====
ADD NEW STUDENT
=====
Enter Matric Number (or 'B' to go back): >
Error: Matric Number cannot be empty.
Enter Matric Number (or 'B' to go back): > B032310583
Enter Student Name: > LOW XIN YEE
Enter Phone Number: > 0173269844
Enter Email: > B032310583@student.utem.edu.my
Enter Faculty: > FTMK

```

## STAFF MENU – MANAGE STUDENT (ADD)

```

=====
CONFIRM STUDENT DETAILS
=====
Matric Number: B032310583
Name: LOW XIN YEE
Phone: 0173269844
Email: B032310583@student.utem.edu.my
Faculty: FTMK
=====
Confirm adding this student? (Y/N): > y

Student added successfully!

Press any key to continue...

```

## STAFF MENU – MANAGE STUDENT (ADD CONFIRMATION)

```

=====
STUDENT RECORDS
=====
Total Records: 8 | Sort by: No_Matric ASC | Page 1 of 2

| Matric No | Name | Email | Phone | Faculty |
|-----|-----|-----|-----|-----|
| B032220056 | HWAM NOORELDEEN ABDALLA ALTOOM | hnam@gmail.com | 123456789 | FTMK |
| B032310202 | MUHAMMAD HAZIQ BIN MOHD AZHAR | haziq@gmail.com | 172467277 | FTMK |
| B032310215 | MUHAMMAD HANIFF BIN MOHD NIZAM | hanif@gmail.com | 123456342 | FTMK |
| B032310420 | MUHAMMAD AKHMAL BIN ROSLAN | akhmal@yahoo.com | 138782670 | FTMK |
| B032310583 | LOW XIN YEE | B032310583@student.utem.edu.my | 173269844 | FTMK |
|-----|-----|-----|-----|-----|

Options:
[</>] Previous/Next Page
[S] Search [O] Sort [R] Reset Filters
[B] Back to Menu
Enter choice: |

```

## STAFF MENU – MANAGE STUDENT (VIEW)

```
=====
                        DELETE STUDENT
=====
Enter Matric Number to delete (or 'B' to go back):>
Error: Matric Number cannot be empty.
Enter Matric Number to delete (or 'B' to go back):> B032310215

                        Student Details:
                        Matric Number: B032310215
                        Name: MUHAMMAD HANIFF BIN MOHD NIZAM

Are you sure you want to delete this student? (Y/N):> y

                        Student deleted successfully!

                        Press any key to continue...
```

### STAFF MENU – MANAGE STUDENT (DELETE)

## STUDENT

```

=====
                        Student Login
=====

Enter Matric Number (or 'B' to go back):> B032310741|

```

## STUDENT LOGIN

```

=====
                        STUDENT PORTAL
=====

Welcome, Riswan bin Hamua

1. View Available Books
2. View My Borrowed Books
3. Check Borrowing Limits
4. Search Transaction History
5. Check Fines
6. Back to Main Menu

=====
Enter your choice: |

```

## STUDENT MENU

```

=====
                        AVAILABLE BOOKS
=====

Total Available Books: 6 | Sort by: Title ASC | Page 1 of 2

| ISBN          | Title                | Author          | Call Number          | Class | Qty |
|-----|-----|-----|-----|-----|-----|
| 9789674816513 | Berani Speaking      | Coach Ridz     | PE 297.571 .R539 2021 os | PE    | 1  |
| 9781498784078 | MATLAB               | Ozgun          | QA 297 .093 2019 os    | QA    | 1  |
| 0764507370    | SQL for Dummies      | Allen G. Taylor | QA 76.3 .T39 2001 os   | QA    | 3  |
| 9789670015873 | Surrounded by Idiots | Thomas Erikson  | BF 789.4 .E73 2019 os  | BF    | 1  |
| 9789671692011 | Teruntuk Langit Yang Bercanda | jiwa          | BP 630.2 .J59 2022 os  | BP    | 1  |

Options:
[</>] Previous/Next Page
[S] Search [O] Sort [R] Reset Filters
[B] Back to Menu
Enter choice: |

```

## STUDENT MENU – VIEW AVAILABLE BOOK

```

=====
MY BORROWED BOOKS
=====

```

| Title           | ISBN          | Borrow Date | Due Date   | Status   | Overdue Days |
|-----------------|---------------|-------------|------------|----------|--------------|
| SQL for Dummies | 9780764501050 | 2025-01-20  | 2025-02-03 | Borrowed | -14          |

Press any key to continue...

### STUDENT MENU – VIEW BORROWED BOOKS

```

=====
BORROWING LIMITS
=====

```

Maximum books allowed: 3  
 Currently borrowed: 1  
 Remaining quota: 2

Press any key to continue...

### STUDENT MENU – VIEW BORROWING LIMITS

```

=====
TRANSACTION HISTORY
=====

```

Student: LOW XIN YEE  
 Matric No: B032310583

Page 1 of 1 | Sort by: Borrow\_Date DESC

| Title           | Borrow Date | Due Date   | Return Date | Status   | Fine (RM) |
|-----------------|-------------|------------|-------------|----------|-----------|
| SQL for Dummies | 2025-01-20  | 2025-02-03 | -           | Borrowed | 0.00      |

Total Fines: RM 0.00

Options:  
 [</>] Previous/Next Page | [S] Status Filter | [D] Date Filter  
 [O] Sort | [R] Reset Filters | [B] Back to Menu  
 Enter choice: |

### STUDENT MENU – TRANSACTION HISTORY

```

=====
CHECK FINES
=====

```

| Title | ISBN | Due Date | Overdue Days | Fine (RM) |
|-------|------|----------|--------------|-----------|
|-------|------|----------|--------------|-----------|

You have no fines. All books are returned on time!

Press any key to continue...

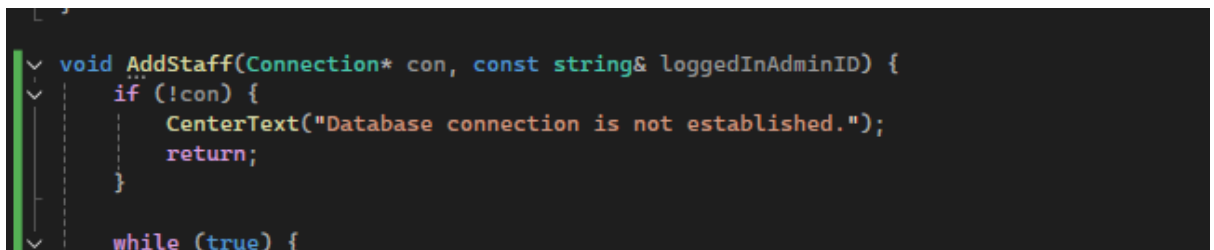
### STUDENT MENU – CHECK FINES

## CHAPTER 4: IMPLEMENTATION

### 4.1 Naming Convention

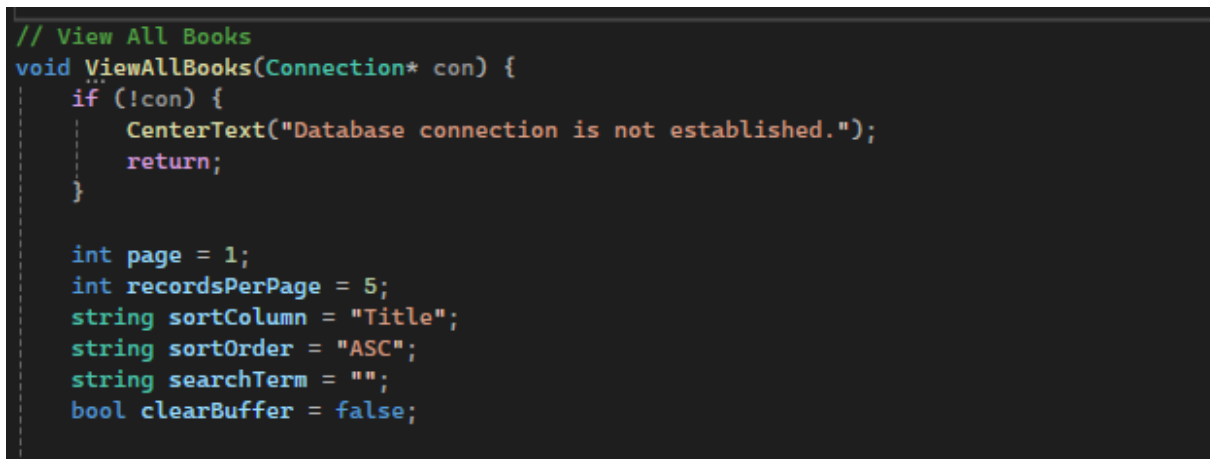
Example of Naming Convention for modules, function, IDs and variables.

Naming Convention ensure the code to be more readable to the programmer.

A screenshot of a code editor showing the implementation of the `AddStaff` function. The code is written in C++ and includes a check for a database connection. The function signature is `void AddStaff(Connection* con, const string& loggedInAdminID) {`. Inside the function, there is an `if (!con) {` block that calls `CenterText("Database connection is not established.");` and then `return;`. Below this, a `while (true) {` loop is partially visible.

```
void AddStaff(Connection* con, const string& loggedInAdminID) {  
    if (!con) {  
        CenterText("Database connection is not established.");  
        return;  
    }  
    while (true) {
```

Figure 4.1.1

A screenshot of a code editor showing the implementation of the `ViewAllBooks` function. The code is written in C++ and includes a check for a database connection. The function signature is `void ViewAllBooks(Connection* con) {`. Inside the function, there is an `if (!con) {` block that calls `CenterText("Database connection is not established.");` and then `return;`. Below this, several variables are initialized: `int page = 1;`, `int recordsPerPage = 5;`, `string sortColumn = "Title";`, `string sortOrder = "ASC";`, `string searchTerm = "";`, and `bool clearBuffer = false;`.

```
// View All Books  
void ViewAllBooks(Connection* con) {  
    if (!con) {  
        CenterText("Database connection is not established.");  
        return;  
    }  
  
    int page = 1;  
    int recordsPerPage = 5;  
    string sortColumn = "Title";  
    string sortOrder = "ASC";  
    string searchTerm = "";  
    bool clearBuffer = false;
```

Figure 4.1.2

## 4.2 Function

Function avoid repetition when writing the code, programmer can just do it once and call it back when needed.

```
// Function to center text dynamically based on console width
void CenterText(const string& text, bool newLine = true) {
    int width = GetConsoleWidth();
    int padding = static_cast<int>((width - text.length()) / 2);
    cout << string(padding, ' ') << text;
    if (newLine) cout << endl;
}
```

Figure 4.2.1

```
void RegisterFirstAdmin(Connection* con) {
    string Admin_ID, password, fullName, email, phoneNumber;

    ClearScreen();

    // display centered headers
    CenterText("=====");
    CenterText("Library Management System");
    CenterText("=====");
    CenterText("Welcome! The system requires an admin.");
    CenterText(" Please register your admin details to proceed.");
    CenterText("=====");

    // Step 1: Admin ID Validation
    do {
        cout << endl; // Add space before input
        CenterAlignedInput("Step 1: Enter Admin ID (e.g., ADM001):", Admin_ID);
    } while (Admin_ID.empty());
}
```

Figure 4.2.2

## 4.3 Array

Array is used to store collection of data such as list of burgers and customers.

```
string join(const vector<string>& vec, const string& separator) {
    if (vec.empty()) return "";
    string result;
    for (size_t i = 0; i < vec.size() - 1; ++i) {
        result += vec[i] + separator;
    }
    result += vec.back();
    return result;
}
```

Figure 4.3.1

## 4.4 Selection

Selection are used to make decisions based on the conditions.

```

    cin >> choice;

    switch (choice) {
    case 1:
        StaffManagementMenu(con, adminID);
        break;
    case 2:
        BookManagementMenu(con, adminID);
        break;
    case 3:
        ReportsMenu(con);
        break;
    case 4:
        CenterText("Logging out... Thank you, " + adminName + "!");
        this_thread::sleep_for(chrono::seconds(2));
        return;
    default:
        CenterText("Invalid choice. Please try again.");
        this_thread::sleep_for(chrono::seconds(2));
    }
} while (true);
}

```

Figure 4.4.1

## 4.5 Control

The code validates user input to ensure correctness and prevent crashes.

```

// Get Quantity
do {
    cout << endl;
    CenterAlignedInput("Enter Quantity (1-10): ", quantity);
    if (cin.fail() || quantity < 1 || quantity > 10) {
        cin.clear();
        cin.ignore(numeric_limits<streamsize>::max(), '\n');
        AlignedErrorMessage("Error: Invalid quantity.", 40);
        continue;
    }
    break;
} while (true);

```

Figure 4.5.1

## Pointer

Pointer are used to dynamically allocate the memory and manage database connections.

```

if (toupper(confirm[0]) == 'Y') {
    PreparedStatement* pstmt = con->prepareStatement(
        "INSERT INTO book (Title, ISBN, Floor_No, Author_Name, Quantity, "
        "Classification_Code, Subject_Number, Cutter_Number, PublicationYear, "
        "CollectionSymbol, Call_Number) VALUES (?, ?, ?, ?, ?, ?, ?, ?, ?, ?, ?)"
    );

    pstmt->setString(1, title);
    pstmt->setString(2, isbn);
    pstmt->setInt(3, floorNo);
    pstmt->setString(4, authorName);
    pstmt->setInt(5, quantity);
    pstmt->setString(6, classificationCode);
    pstmt->setString(7, subjectNumber);
    pstmt->setString(8, cutterNumber);
    pstmt->setInt(9, publicationYear);
    pstmt->setString(10, collectionSymbol);
    pstmt->setString(11, callNumber);

    pstmt->executeUpdate();
    delete pstmt;
}

```

Figure 4.6.1

## 4.6 Error Handling

Error handling are used to notify the error in the system such as connection to database error or when the user enter a wrong format for anything

```

}
catch (SQLException& e) {
    CenterText("Error: Failed to add book.");
    CenterText("SQL Error: " + string(e.what()));
    cout << endl;
    CenterText("Press any key to continue...");
    getch();
    return;
}
}
}

```

Figure 4.7.1

```

CenterText("=====");
CenterText("Library Management System");
CenterText("=====");
CenterText("Welcome! The system requires an admin.");
CenterText(" Please register your admin details to proceed.");
CenterText("=====");

// Step 1: Admin ID Validation
do {
    cout << endl; // Add space before input
    CenterAlignedInput("Step 1: Enter Admin ID (e.g., ADM001):", Admin_ID);

    if (Admin_ID.empty()) {
        AlignedErrorMessage("Error: Admin ID cannot be empty. Please re-enter.", -40);
    }
    else if (!isValidUserID(Admin_ID, "ADM")) {
        // Print the error message directly below
        AlignedErrorMessage("Error: Invalid ID format. Use 'ADM' followed by 3-5 digits (e.g., ADM001).", -40);
    }
}

```

Figure 4.7.2

## **CHAPTER 5: CONCLUSION**

### **5.1 Constraints**

Library Management System face several constraints that need careful consideration for its successful implementation and ongoing operation. Firstly, Integrating real-time updates across multiple modules (e.g., borrowing, returns, fines) requires meticulous handling of database transactions. Inconsistent synchronization could lead to data discrepancies, such as books appearing available when they are checked out. Secondly, the system must efficiently handle high traffic during peak periods, such as exam seasons or enrollment weeks. Slowdowns or crashes during these times could disrupt borrowing/returning processes, leading to user frustration and operational bottlenecks. Lastly, Regular updates are required to address security flaws and improve features. Limited technical resources might delay critical updates, leaving the system vulnerable to exploits or obsolescence.

### **5.2 Future Improvements**

For future improvements, Library Management System must prioritize Integrating AI-Powered Features, such as personalized book recommendations and demand forecasting, will modernize operations and improve user engagement by aligning resources with borrowing patterns. Secondly, Expanding Mobile Accessibility through a dedicated app will enable members to browse catalogs, renew loans, and pay fines remotely. Features like QR code scanning and push notifications will streamline interactions and enhance convenience. Lastly, Library Management System could implementing Cloud-Based Scalability through elastic resource allocation and optimized database architectures will ensure seamless performance during peak usage, such as exam seasons or enrollment periods, preventing system slowdowns or outages.

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